Singapore School Kinderland
Secondary-1
Computer Science
(Chapter 1: Sections 1.1 to 1.1.3)
Term 1

Class Test-1 (Sections 1.1 to 1.1.3: Number Systems & Hexadecimal)

Date: _____ **Name: _____**

Total marks: 15 **Marks obtained: _____**

Teacher's signature: _____ **Parent's signature: _____**

Question 1: The Binary Number System & Denary Conversion [Total: 4 marks]

Textbook Reference: Section 1.1.1 & 1.1.2 (Pages 2–7)

(a) Computers only process binary data (0s and 1s). Explain why computers use the binary number system to represent and process data. **[1 mark]**

(b) Convert the following 8-bit binary number into a denary number. Show your working:

0 1 1 0 1 0 1 1 **[1 mark]**

(c) Convert the denary number 174 into an 8-bit binary number using successive division by 2 or place-value subtraction. Show all working steps clearly. **[2 marks]**

Question 2: Hexadecimal Number Conversions [Total: 4 marks]

Textbook Reference: Section 1.1.2 (Pages 8–12)

(a) Convert the 12-bit binary number 0111 1101 0100 into Hexadecimal notation. Show your working. **[1 mark]**

(b) Convert the hexadecimal value 5B8 into an unsigned binary number. **[1 mark]**

(c) Convert the hexadecimal number 2C5 into a denary number. Show your full calculation. **[2 marks]**

Question 3: Denary to Hexadecimal Conversion **[Total: 3 marks]**

Textbook Reference: Section 1.1.2 (Pages 11–12)

(a) Convert the denary number 490 into Hexadecimal. Show your successive division by 16 method and calculations clearly. **[2 marks]**

(b) State how many hexadecimal digits are needed to represent the denary value 490. **[1 mark]**

Question 4: Uses of Hexadecimal & HTML Colour Codes **[Total: 4 marks]**

Textbook Reference: Section 1.1.3 (Pages 13–15)

(a) State two reasons why software developers and engineers prefer to use hexadecimal notation instead of pure binary notation. **[2 marks]**

(b) (i) In web design, colours are defined using 6-character hexadecimal codes in HTML (#RRGGBB). State which colour is represented by the code #FF0000. **[1 mark]**

(b) (ii) Explain what the hexadecimal colour codes #000000 and #FFFFFF represent in terms of RGB light intensity. **[1 mark]**